Team effectiveness is a context problem, not a talent problem

The research on what makes engineering teams effective has been settled for a decade. Most engineering leaders are still fixing the wrong layer.

Google spent years studying about 180 of its own teams across 250+ attributes, expecting to find the winning combination of talent — the right seniority mix, the right personalities, the right people. What they found instead: who was on the team mattered less than how the team worked. Group norms beat individual attributes, every time.

That's Project Aristotle. It ranked five dynamics of effective teams: psychological safety first, by a significant margin; then dependability, structure and clarity, meaning, and impact. Teams where people could admit mistakes, ask questions, and take interpersonal risks without getting punished were rated effective about twice as often. In Google's sales data, high-safety teams beat their targets by about 17% on average while low-safety teams missed by about 19%. Those are Google's numbers from its own org — the point isn't the exact figure, it's that the effect shows up in hard revenue terms, not just survey vibes.

The same answer arrives from a completely different field. Ron Westrum, studying aviation and medicine, classified organizations by how they treat information: pathological (power first, messengers get shot), bureaucratic (rules first, messengers get ignored), generative (performance first, messengers get trained). DORA's research on software teams found generative cultures correlate with roughly 30% higher organizational performance — one analysis, correlational, but it's the same signal from an independent direction.

Two research programs, thirty years apart, same conclusion: **the mechanism is information flow.** Psychological safety is the team-level name for it. Generative culture is the org-level name for it. Both describe the same thing: can the person who sees the problem say it out loud without becoming the problem?

This is why so many improvement efforts don't stick. You adopt the practices — the retros, the design reviews, the incident process — and six months later the team is exactly the same. The practices aren't the problem, and they're not the solution. If the context punishes honesty, the retro becomes a meeting where nobody says the thing. The process works in one team and dies in the next, and the difference is never the process. It's whether the people running it are safe.

Every factor people argue about — leadership, team structure, incentives, communication, ownership — resolves into the same question: does information flow to the person who can act on it, and is it safe for that person to say what they see? Structure debates are usually information-flow debates wearing org charts. Incentive debates are usually information-flow debates wearing compensation spreadsheets. Leadership debates are usually information-flow debates wearing performance reviews.

And here's the part that changed this year. DORA's 2025 report — about 5,000 respondents, roughly 90% using AI — found that AI is an amplifier, not an equalizer. It magnifies existing strengths and existing weaknesses. Teams with strong foundations got stronger; teams with weak context got weaker. Adoption alone didn't predict outcomes; the surrounding system did. The same tools, the same models, and the gap between teams widened instead of narrowed.

That means the context question is no longer a soft HR concern. It's the thing that decides whether your AI investment compounds or corrodes. DORA's own capability model is basically a list of context conditions — a clear AI stance, healthy data ecosystems, AI-accessible internal data, strong version control, small batches, user-centric focus, solid internal platforms. None of those are "hire better people." All of them are "make the environment the good behavior can survive in." The teams that already had those conditions got the gains. The teams that didn't got more of what they already had.

So what do you actually check? Not the org chart, not the process maturity score. Six signals, five minutes:

1. **When someone doesn't know something, do they ask — or fake it?** Faking means knowledge gaps stay hidden and compound in silence.
2. **When bad news exists, how fast does it reach the person who can act on it?** Delayed bad news is the universal failure mode; the org chart is how fast the truth travels.
3. **When something fails, is the first response inquiry or blame?** Westrum's whole model fits in this one question. Inquiry gets the next fix; blame gets the next cover-up.
4. **Do incentives reward the behavior the team needs, or the behavior that looks good in review?** The metric that rewards shipping broken things wins, every time, regardless of what the charter says.
5. **Does the team own outcomes, or just tickets?** Ticket owners optimize for done; outcome owners optimize for works.
6. **Are mistakes cheap enough to make early and often?** If the first mistake is expensive, nobody makes the second one — they just hide the first.

The first three are the information-flow layer. If those are failing, no practice adoption fixes anything — you're polishing a process in a culture that punishes the behavior the process depends on. The last three are the alignment layer: the flow can be perfect and the system still wastes it if incentives point elsewhere, ownership stops at the ticket, or mistakes are expensive enough that nobody takes risks. Different failures, different fixes. Most teams I've seen are failing on 1–3 and running retros as if they were failing on 4–6.

One honest bound. Context is not fair, and it does not protect individuals — the organization protects itself. You can be the most effective engineer on a team whose context punishes honesty, and the context wins. That's not a reason to stop trying; it's a reason to stop pretending the fix is one more practice. Fix the context and the practices follow. Fix the practices and the context stays.

That's the whole difference between teams that deliver and teams that struggle — and it was true before AI. AI just made the gap visible.
